

RAG Pipeline Integration with Vector Databases

Integrating vector databases into RAG pipelines involves several coordinated steps. First, documents are chunked into appropriately-sized segments, typically 256-1024 tokens, balancing context completeness and retrieval granularity. Chunk overlap preserves context across boundaries. Metadata is extracted and associated with chunks—source document, page numbers, timestamps, categories, or custom fields enabling filtered retrieval. Embedding models process chunks, converting text to vectors. Batch processing improves efficiency by embedding multiple documents simultaneously. Embedding vectors, chunk content, and metadata are inserted into the vector database. The database creates indexes enabling fast similarity search. When a user query arrives, it's embedded using the same embedding model as the documents. The embedded query is searched against the vector database, retrieving top-k most similar chunks. Metadata filtering can restrict search to specific documents or time ranges. Retrieved chunks are ranked, either maintaining database ranking or applying cross-encoder reranking. The top-ranked chunks are formatted into a context prompt. The language model processes the query with retrieved context, generating an informed response. Complete pipelines handle streaming results, timeout management, and error recovery.